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Institutional trust as a critical determinant of households' willingness to pay for improved water services in Leku Town, Ethiopia

Wogene Markos Dumo^{1*}

*Correspondence:
Wogene Markos Dumo
wogister@gmail.com

¹Department of Economics,
Hawassa University, Hawassa,
Ethiopia

Abstract

Universal access to safe water remains a critical challenge in the Global South. This study examines households' willingness to pay (WTP) for improved water services in the secondary town of Leku Town, Ethiopia, identifying key socioeconomic, demographic, and institutional determinants. Using a cross-sectional contingent valuation survey with stratified random sampling of 193 households and binary logistic regression analysis, it is found that 68.9% of households were willing to pay for improvements. Household income was the strongest positive predictor (Adjusted Odds Ratio, AOR=7.49, $p < 0.001$). Crucially, institutional factors demonstrated profound negative effects. A negative attitude toward the water provider (AOR=0.19, $p < 0.05$) and dissatisfaction with service timing (AOR=0.17, $p < 0.01$) each reduced WTP odds by over 80%. Female respondents showed significantly higher WTP than males (AOR=2.08 for females, $p < 0.05$). The findings demonstrate that WTP depends not only on economic capacity but critically on institutional trust. The study concludes that securing sustainable financing requires both pro-poor pricing and targeted strategies to rebuild institutional credibility. This paper recommends implementing pro-poor tariff structures, executing visible service improvements to rebuild trust, fostering transparent communication, and ensuring inclusive community engagement, particularly of women, to secure sustainable water service financing.

Keywords Willingness to pay, Contingent valuation, Improved water services, Binary logistic regression, Institutional trust, Service quality, Ethiopia, Secondary towns

1 Introduction

Access to safe, reliable, and affordable drinking water is recognized as a fundamental human right [71] and central to the Sustainable Development Goals, notably SDG 6 [72]. Yet, disparities remain unambiguous: nearly two billion people still lack managed drinking water, with sub-Saharan Africa and fragile contexts most affected [18, 82]. Deficient infrastructure and governance perpetuate poverty, gender inequality, and public health crises, as waterborne diseases such as diarrhoea, cholera, and typhoid continue to cause high child mortality [55, 84]. Moreover, the time burden of water collection, borne



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largely by women and girls, constrains education and productivity, reinforcing intergenerational poverty [27, 62].

Ethiopia has prioritized water in its national development agenda [22], with policies such as the Water Sector Policy and One WASH Program outlining strategies for expanded access [47, 51]. National coverage has improved [81], but aggregate figures mask regional inequalities, rural-urban divides, and persistent issues of reliability and quality [4, 26]. Many households counted as “served” face intermittent supply, long queues, and unsafe alternatives [14, 63].

Challenges are most acute in secondary towns (pop. 20,000–100,000), which face rapid urbanization and strained infrastructure [15, 70]. Local water utilities often lack resources for expansion and maintenance, leading to poor service, eroded trust, and suppressed revenue collection, a cycle that undermines financial sustainability [3, 74]. Addressing this requires shifting from supply-driven projects to models integrating sustainable financing, demand responsiveness, and governance reform. This study, conducted in Leku Town (pop. ~31,700), responds to this gap, recognizing that institutional trust factors are also relevant for sanitation financing.

In Leku, intermittent piped supply forces reliance on unimproved sources. Here, economic valuation methods such as the Contingent Valuation Method (CVM) are indispensable for evidence-based policy [23]. By eliciting households’ maximum Willingness to Pay (WTP), CVM informs tariff design, project viability, and subsidy mechanisms [10, 80]. Ethiopian CVM studies consistently highlight socioeconomic determinants: income [26, 43], education [46], female headship [65], and time spent fetching water [33].

However, two gaps persist. First, secondary towns like Leku remain under-researched despite their critical role in Ethiopia’s urbanization trajectory [21, 67]. Second, institutional and perceptual factors are underexplored. Research shows that trust in service providers profoundly shapes WTP [49, 52]. In contexts of weak regulation and poor past delivery, households’ willingness to contribute depends on confidence in water utility performance [44, 56]. Ignoring this “trust deficit” undermines cooperation [68]. Evidence from solid waste management [60] and ecosystem services [2] confirms institutional trust as a key mediator of WTP. This study is further informed by theories of institutional trust and the social contract in public service delivery [58, 73]. These theories posit that citizens’ compliance and financial contributions are contingent upon their belief in the reliability, competence, and fairness of public institutions. A breakdown in this perceived contract, where services are unreliable or unaccountable, erodes trust and suppresses the willingness to cooperate, including through tariff payments [68].

This study’s novelty lies in two contributions: (1) addressing the contextual gap by focusing on Leku Town, and (2) integrating institutional and perceptual variables specifically households’ attitudes toward the water utility and satisfaction with service performance into households’ attitudes toward the water utility and satisfaction with service performance, into WTP analysis. Its objectives are:

1. To identify socioeconomic, demographic, and institutional determinants of WTP for improved water services in Leku Town.
2. To elicit and quantify mean WTP for proposed service improvements through a hypothetical market scenario.

While focused on water supply, the findings also inform integrated WASH approaches, with institutional trust factors relevant for sanitation financing.

2 Literature review and conceptual framework

2.1 Theoretical foundations of willingness to pay

Willingness to Pay (WTP), rooted in welfare economics and the valuation of non-market goods [6], denotes the maximum monetary amount individuals would sacrifice to secure an improvement in a good or service, or the minimum compensation they would accept for its loss (Willingness to Accept, WTA) [23]. For public goods such as improved water services, WTP provides a proxy for the economic value of benefits like enhanced health, time savings, and convenience [29].

The Contingent Valuation Method (CVM) remains the most widely applied stated-preference technique for eliciting WTP. By constructing a hypothetical market and presenting respondents with a detailed scenario, CVM directly captures valuation [11]. While concerns persist regarding hypothetical, strategic, and starting point biases, well-designed CVM studies are now regarded as robust tools for policy analysis [37]. Best practices outlined by the National Oceanic and Atmospheric Administration (NOAA) panel [1], including credible scenarios, pre-testing, referendum-style formats, and rigorous econometric analysis, have been successfully applied in numerous water valuation studies across the developing world [77, 79].

2.2 Empirical evidence on determinants of WTP for water services

A comprehensive review of the global and Ethiopian literature reveals a consistent set of variables that influence WTP for improved water services. These can be categorized into economic, demographic, and situational factors.

2.2.1 Economic factors

- *Income*: This is arguably the most robust and universally significant determinant. Economic theory posits that the ability to pay is a prerequisite for stated WTP. Numerous studies in Ethiopia [7, 26] and elsewhere [12, 30] confirm a strong positive correlation between household income and WTP. Higher-income households have a greater capacity to absorb tariff increases and often place a higher value on convenience and reliability.
- *Wealth and Assets*: Beyond cash income, household wealth, proxied by ownership of assets like land, livestock, or durable goods, has also been linked to higher WTP, as it reflects a broader economic capacity and resilience [42, 57].

2.2.2 Demographic and social factors

Socioeconomic characteristics of households are consistently examined as predictors of WTP.

- *Education*: The education level of the household head is generally a positive determinant of WTP [34, 46]. Education enhances awareness of the health and economic benefits of improved water and may foster civic responsibility [28].

- *Gender*: Household headship often shows significant, context-specific effects. Evidence from Ethiopia and sub-Saharan Africa indicates female-headed households exhibit higher WTP [48, 65], largely due to women's disproportionate burden in water collection and management [17].
- *Age*: Findings are mixed. Older household heads, with stable livelihoods and longer exposure to scarcity, may be more willing to pay [66], while younger households often show greater openness to change and tariff adjustments [9].
- *Household Size*: The relationship remains ambiguous. Larger households may value improvements more due to higher consumption, yet budget constraints can suppress WTP [31].

2.2.3 Situational and perceptual factors

Household water-related conditions are critical determinants of WTP.

- *Current Source and Cost*: Reliance on costly or inconvenient alternatives, such as private vendors, often increases WTP for reliable public supply, as it represents a direct cost-saving [16, 78].
- *Time Burden*: Greater time spent fetching water strongly motivates WTP, reflecting significant opportunity costs that disproportionately affect women and girls [20, 40].
- *Perceived Quality and Health Risks*: Experiences with waterborne disease or perceptions of poor-quality water drive WTP, as households seek to mitigate health costs and risks [19, 38].

2.3 The critical gap: institutional trust and service performance

Recent literature highlights the importance of institutional and governance factors in shaping WTP. The New Institutional Economics framework argues that transaction costs and institutional arrangements strongly influence economic behavior [83]. Applied to WTP, household decisions depend not only on preferences but also on perceptions of the provider's credibility, competence, and fairness.

Institutional Trust: Defined as belief in another party's reliability and competence [58], trust in public utilities emerges from consistent service delivery, transparency, and accountability [73]. Low trust fosters skepticism, "Why pay for nothing?" and reduces WTP. Empirical studies in China's solid waste management [60] and Kenya's water services [39] confirm trust as a significant, often dominant, predictor of WTP.

Service Satisfaction: Direct experiences with service quality, irregular supply, low pressure, or frequent breakdowns shape perceptions. Dissatisfaction signals institutional failure, lowering expectations of improvement and suppressing WTP [59, 75]. Research on public utilities consistently links service quality to payment compliance and support for tariff adjustments [45].

In Ethiopia's decentralized governance, where secondary-town utilities often face limited oversight and resources [5, 15], trust and satisfaction are particularly critical. Yet, few studies explicitly model these institutional dimensions alongside traditional socio-economic factors. This study seeks to address that conceptual and empirical gap.

2.4 Conceptual framework

Based on the synthesis of the literature, this study is guided by the conceptual framework presented in Fig. 1, where WTP is a function of three clusters:

1. **Economic Capacity** (Ability to pay, primarily income).
2. **Demographic/Social Characteristics** (Age, gender, education, etc.).
3. **Institutional/Perceptual Factors** (Attitude, satisfaction, i.e., Willingness to pay, mediated by trust).

The framework is informed by the consumer behavior paradigm, where purchase decisions require both *ability* and *willingness* [53], adapted here to public good valuation. The core hypothesis of this framework is that both *ability* (economic capacity) and *willingness* (driven by trust and satisfaction) are necessary conditions for a positive WTP decision. A deficiency in either can lead to an unwillingness to pay, even if the other condition is met.

3 Methods

3.1 Study area description

This research was conducted in Leku Town, which is situated in the Sidama National Regional State (SNRS), Ethiopia. The town holds a strategic position approximately 23.5 km south of Hawassa, the vibrant capital city and economic hub of the SNRS. Leku serves as a crucial administrative and commercial center for the surrounding rural kebeles, acting as a market town for agricultural produce. The town has an estimated 31,700 total population according to the Leku town administration (2023/24). These inhabitants are distributed across an estimated 6,604 households, which are administratively organized into five distinct kebeles (Leku Mekakele, Leku Afecho, Leku Harena, Leku Hidase, and Leku Mencha).

The choice of Leku Town as a case study is strategic and purposeful. It embodies the quintessential characteristics of many emerging secondary urban centers across Ethiopia that are experiencing the twin pressures of rapid population growth and increasing strain on existing public infrastructure, particularly water supply systems [15, 85]. Like many similar towns, Leku's water supply is characterized by intermittency, with many residents receiving piped water for only a few hours every few days. This forces households to supplement their supply from unimproved sources such as shallow wells,

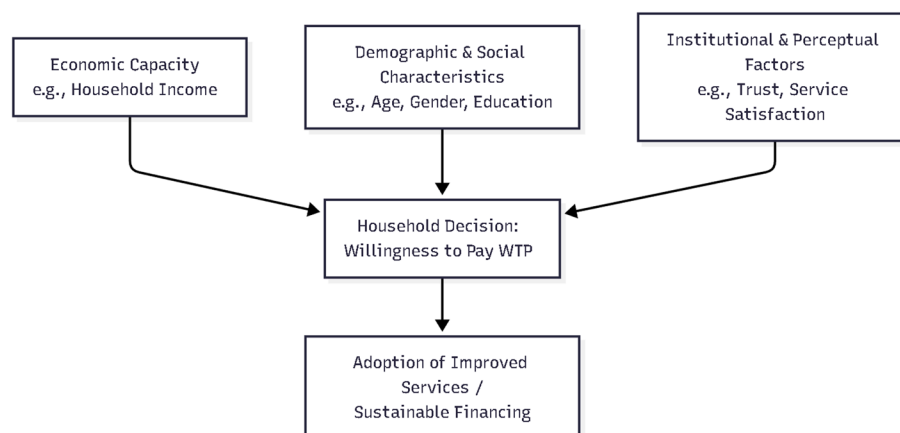


Fig. 1 Conceptual Framework of Household WTP for Improved Water Services

springs, or expensive private water vendors, mirroring a common pattern of coping strategies observed in under-serviced urban areas [41].

3.2 Research design and sampling technique

A cross-sectional study design was employed, collecting data on willingness to pay (WTP) and explanatory variables at a single point in time. This design is standard for establishing associative relationships in contingent valuation studies within development contexts [79].

Stratified random sampling was used to ensure a representative sample across the town's socio-spatial diversity. The five administrative kebeles of Leku Town serve as strata to capture variations in water access, settlement patterns, and socioeconomic status.

3.2.1 Population and household estimation

To establish the sampling frame, the most recent verified population figure of 31,700 was obtained from the Leku Town Administration (2023/24). Household estimates were derived by dividing this population by the average household size. The Central Statistical Agency [13] reported an average household size of 4.8 for the Sidama region, while our preliminary survey observations indicated 4.7 persons per household in Leku. Using the locally observed figure of 4.7 yields approximately 6,745 households, which aligns with the administrative records and local planning data.

3.2.2 Sample size calculation

The sample size was calculated using Yamane's [86] formula for a finite population:

$$n = \frac{N}{1 + N(e)^2}$$
 where N is the number of households (6,745) and e is the margin of error. A margin of error of 7% was selected to balance statistical precision with practical resource constraints in this town-level exploratory study. While 5% is conventional in many disciplines, wider margins are acceptable and often necessary in small-scale, resource-constrained contingent valuation studies where the primary goal is to identify key determinants and provide policy-relevant estimates rather than achieve national-level precision consistent with practical challenges noted by Whittington [80]. Using the formula:

$$n = \frac{6,745}{1 + 6,745(0.07)^2} \approx 198$$

3.2.3 Sampling procedure

The target sample of 198 households was allocated proportionally to each kebele based on its share of total households. Within each stratum, households were selected using simple random sampling from updated household lists obtained from the town administration and community leaders.

3.2.4 Field implementation and response rate

A final sample of 193 households was successfully surveyed, achieving a completion rate of 97.5% against the target. This yields a sampling fraction of 2.9% and an effective margin of error of 7.1%. All contacted households eventually participated after multiple scheduled revisits, resulting in a de facto 100% response rate for the contacted sample. This high engagement rate is attributed to: (1) the acute community concern over water

issues, (2) locally-recruited, native-speaking enumerators who understood community norms, (3) explicit support from kebele administrators, and (4) a persistent 'three-visit protocol' implemented for each selected household. Such high response rates, while unusual in large-scale surveys, are achievable in focused community-based studies with strong local partnerships and persistent fieldwork (e.g., [21]).

The interview was conducted with the household head or, in their absence, another adult (≥ 18 years) primarily responsible for water management.

3.3 Data types, sources, and collection methods

The study relied exclusively on primary cross-sectional data collected specifically for this research purpose. The main instrument for data collection was a structured questionnaire, which was developed after an extensive and critical review of relevant literature on contingent valuation methods and water service studies in Ethiopia and similar contexts [7, 26, 33, 76].

The questionnaire underwent a rigorous development process. It was initially prepared in English, translated into Sidamigna (the official working language of the region and the primary language of the respondents), and then back-translated into English by an independent translator to check for conceptual consistency and accuracy. The instrument was pre-tested on 15 households in a kebele not included in the final sample to assess the clarity, flow, logical sequence, and most importantly, the understanding and credibility of the contingent valuation scenario. Necessary adjustments to wording, bid amounts, and scenario description were made based on the pre-test feedback.

The final questionnaire comprised four main sections:

1. *Socio-Demographic and Economic Characteristics*: This section gathered data on variables such as the age, gender, marital status, and education level of the respondent, as well as household size, occupation, and household income. Income was measured categorically to reduce respondent reluctance, asking for the range of total monthly household income from all sources.
2. *Current Water Supply and Use Patterns*: This section collected detailed information on the primary water source for drinking and other domestic purposes, round-trip distance to the source, time spent collecting water per day, average quantity of water used, perceived water quality (on a Likert scale), self-reported reliability of supply, and the occurrence of water-related illnesses in the household in the past six months.
3. *Contingent Valuation Method (CVM) Module*: This was the core. It began with a detailed, neutral description of the current water service situation, followed by a precise definition of the "improved service" being valued (24/7 reliable access to clean piped water, reduced disease, professional service). The hypothetical payment scenario was framed within the context of the existing monthly water bill, using an increase in this bill as the payment vehicle to enhance realism. To mitigate potential starting-point bias, five bid amounts (50, 75, 100, 125, 150 ETB) were selected based on current tariffs and pilot survey results, and were randomly assigned to respondents. To reduce hypothetical bias, enumerators were trained to present the scenario in a neutral, concrete manner, and a 'cheap talk' script was employed, reminding respondents to answer as if real monetary commitment was required [1]. A single-bounded dichotomous choice (SBDC) format was used.

4. *Perceptions and Institutional Trust:* This section assessed households' overall attitude towards the current water utility on a scale from very negative to very positive. It also gauged their level of satisfaction with specific service attributes, particularly the timing and reliability of water provision, using a Likert scale.

Data collection was conducted over a period of six weeks. Eight trained enumerators, who were fluent in the local language (Sidama Afoo), administered the questionnaires through face-to-face interviews. This method was chosen to ensure high response rates, clarify any ambiguous questions, and minimize misunderstandings, particularly regarding the hypothetical market scenario. A 100% response rate was achieved through enumerators making multiple revisit attempts. Informed consent was obtained verbally from all participants before commencing the interview.

3.4 Methods of data analysis

Collected data were checked for completeness and logical consistency (e.g., time spent vs. source distance). Double data entry was performed in SPSS (Version 27) for validation. SPSS was used for descriptive statistics and data management. Stata 17 was used for binary logistic regression and advanced model diagnostics (VIF, Hosmer-Lemeshow), leveraging its specialized econometric capabilities. The analysis proceeded in two stages:

1. *Descriptive Statistics:* This was used to summarize and present the sociodemographic and economic characteristics of the sampled households, their current water use patterns, and the overall distribution of WTP responses. Results were presented using frequencies, percentages, means, and standard deviations in the form of tables and figures.
2. *Econometric Model:* Since the dependent variable (Willingness to Pay) is dichotomous (coded as 1 for 'Yes' and 0 for 'No'), a binary logistic regression model was employed to identify the factors influencing households' WTP. The logistic model is preferred in such cases because it does not assume a linear relationship between the dependent and independent variables, it constrains the predicted probabilities between 0 and 1, and it handles the binary nature of the outcome effectively (Hosmer, Lemeshow, and Sturdivant 2013).

The binary logistic regression model is specified as:

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k)}}$$

Where:

- $P(Y = 1)$ is the probability that a household is willing to pay for improved water services.
- β_0 is the intercept.
- $\beta_1, \beta_2, \dots, \beta_k$ are the coefficients of the explanatory variables $X_1, X_2, \dots, X_k$

The independent variables included in the initial model were selected based on economic theory and the empirical literature reviewed earlier. They were: household income (ordinal), age of respondent (ordinal), gender (dummy: 1 = Female), marital status (dummy: 1 = Unmarried), education level (ordinal), household size (continuous), current quantity of water used (continuous), attitude towards the water utility

(dummy: 1 = Negative), and satisfaction with the time-of-service provision (dummy: 1 = Dissatisfied), and primary water source (dummy: 1 = Piped).

3. *Model Diagnostics*: To ensure the robustness and validity of the regression results, standard model diagnostics were performed. Multicollinearity among the independent variables was checked using the Variance Inflation Factor (VIF); a VIF value greater than 10 was taken as an indication of serious multicollinearity [50]. The overall goodness-of-fit of the model was assessed using the Omnibus Tests of Model Coefficients and the Nagelkerke R^2 statistic. The Hosmer-Lemeshow test was used to evaluate how well the model's predictions match the observed data; a non-significant chi-square value ($p > 0.05$) indicates a good fit [32].

4 Results

4.1 Descriptive analysis: sociodemographic profile and WTP

The survey successfully collected data from all 193 sampled households, yielding a 100% response rate. The sociodemographic characteristics of the respondents are summarized in Table 1. The sample exhibited a balanced gender distribution with a slight majority of female respondents (53.9%). A majority of the respondents (53.9%) were above 36 years of age, and most were married (73.6%). In terms of educational attainment, the largest group held a Diploma certificate (36.3%), followed by those who had completed primary education (26.9%). The average household size was 4.7 persons ($SD = 1.8$). Household income distribution shows that most households (34.7%) fell into the 2001–4000 ETB monthly income bracket, indicating a generally low-income community.

Table 1 Sociodemographic characteristics of respondents ($N = 193$). Source Author's computation

Characteristic	Category	Frequency	Percentage (%)
Gender	Female	104	53.9
	Male	89	46.1
Age Group	18–25 years	32	16.6
	26–35 years	57	29.5
	Above 36 years	104	53.9
Marital Status	Married	142	73.6
	Unmarried	51	26.4
Education Level	No formal education	38	19.7
	Primary (1–8)	52	26.9
	Secondary (9–12)	33	17.1
	Diploma and above	70	36.3
Average Monthly Income (ETB)	< 2000	45	23.3
	2001–4000	67	34.7
	4001–6000	54	28.0
	> 6000	27	14.0
Avg. Daily Water Use (Liters)	Mean (Standard Deviation)	42.3	(18.7)
Attitude of households towards water Utility	Negative	68	35.2%
	Neutral/Positive	125	64.8%
Satisfaction with Timing	Dissatisfied	121	62.7%
	Neutral/Satisfied	72	37.3%
Types of illness (reported case)	Diarrhea	118	85%
	Typhoid	44	32%

Negative attitude towards water utility: Dummy variable (1 = Agreed/Strongly Agreed that 'The water utility is inefficient and untrustworthy'). Dissatisfaction with timing: Dummy variable (1 = Dissatisfied/Very Dissatisfied with the regularity and timing of water supply)

The reported average water use of 42.3 L per household per day is notably low, equivalent to less than 10 L per capita per day. This likely reflects the severe constraints of the current intermittent service, where consumption is limited by availability, long queuing times, and storage capacity, rather than by need.

The survey also revealed significant health and time burdens associated with the current water situation. A substantial number of households (72.0%) reported that at least one family member had fallen ill in the past six months from diseases they attributed to using water from unimproved sources. The most commonly cited illnesses were diarrhea (85% of reported cases) and typhoid (32% of reported cases), highlighting the direct health benefits that improved services could deliver. Furthermore, the average time spent collecting water was 52 min per round trip, with women and girls being the primary collectors in 89% of households.

Regarding the primary outcome variable, a significant majority of 133 households (68.9%) expressed a positive willingness to pay for the proposed improved water services, while 60 households (31.1%) were unwilling to pay. The mean WTP, estimated using the Turnbull estimator for single-bounded dichotomous-choice data, was 98.5 ETB per month (95% CI: 86.2–110.8 ETB). This represents approximately 2.5% to 5% of the median monthly household income (2001–4000 ETB) reported in the sample.

Figure 2 above shows how positive WTP responses vary with bid amounts. This suggests a significant potential market and community backing for investing in water improvements. The data plotted displays the typical downward trend seen in Contingent Valuation Method (CVM) surveys.

4.2 Econometric results: determinants of willingness to pay

The binary logistic regression model was executed to identify the factors significantly influencing households' WTP for improved water services. Variables such as education level and household size were included in the initial model based on theoretical considerations. However, they were found to be statistically insignificant ($p > 0.1$) and were excluded in subsequent models to improve parsimony and interpretability. Sensitivity analyses confirmed that their exclusion did not materially affect the main results.

Distribution of Positive WTP Responses Across Bid Amounts (Fig 2)

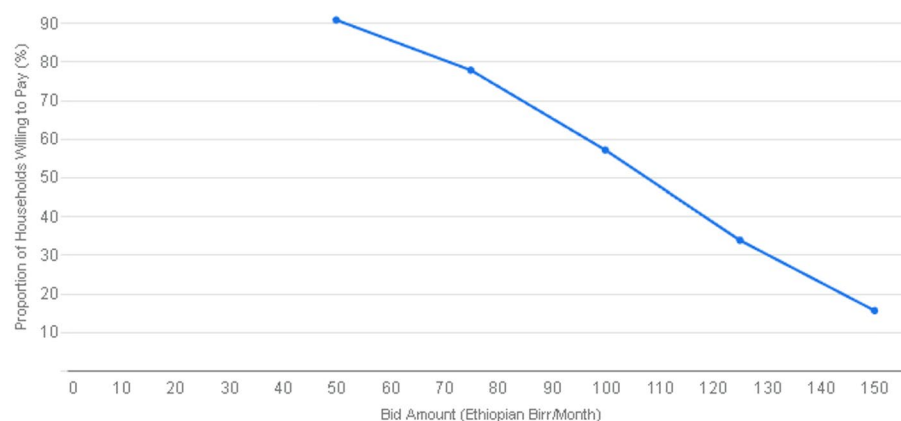


Fig. 2 Distribution of Positive WTP Responses Across Bid Amounts (N = 193)

Diagnostic tests confirmed model robustness. The VIF for all variables ranged from 1.12 to 2.58. The Omnibus Test was significant ($\chi^2 = 145.72$, $p < 0.001$). The Nagelkerke R^2 was 0.968. The Hosmer-Lemeshow test indicated good fit ($\chi^2 = 6.84$, $p = 0.555$). The high Nagelkerke R^2 is likely due to the powerful, dichotomous institutional variables in a model predicting a binary outcome. To validate, a socioeconomic-only model yielded $R^2 = 0.61$. The institutional variables thus provide substantial unique explanatory power. A nested model analysis confirmed this: adding the two institutional variables to a base socioeconomic model increased the Nagelkerke R^2 by 0.358 ($\Delta\chi^2 = 84.3$, $p < 0.001$).

The results of the binary logistic regression analysis are presented in Table 2. The table reports the coefficient (B), the Wald statistic, the significance level (p-value), and the odds ratio [Exp(B)] and 95% Confidence Interval (CI) for each variable that was found to be statistically significant in the final parsimonious model. Variables such as education level and household size were included in the initial model but were found to be statistically insignificant ($p > 0.1$) and were subsequently excluded to enhance model parsimony.

The key findings from the model are as follows:

Household Income: This emerged as the most powerful positive predictor of WTP. The coefficient is positive and highly significant ($p < 0.01$). The odds ratio of 7.490 indicates that for each step increase in income category (e.g., moving from the 2000–4000 ETB bracket to the 4001–6000 ETB bracket), the odds of a household being willing to pay become 7.5 times higher, holding all other factors constant. This strongly confirms that economic capacity is a fundamental driver of WTP.

Age of Respondent: Age had a significant positive effect on WTP ($p < 0.01$). Older respondents were more likely to be willing to pay for improved services, with the odds increasing by a factor of 2.036 for each age category increase.

Gender: Gender was a significant determinant ($p < 0.05$). The positive coefficient for the female category (coded as 1) and an odds ratio of 2.080 indicate that female respondents were about twice as likely as male respondents to express a positive WTP, all else being equal.

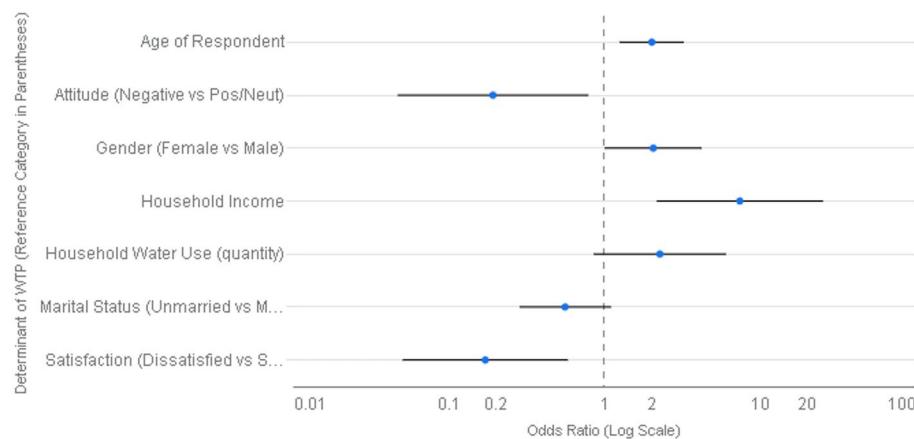
Marital Status: Marital status was significant at the 10% level ($p < 0.1$). The negative coefficient for the unmarried category suggests that unmarried respondents were less likely to be willing to pay compared to their married counterparts (Odds Ratio = 0.564).

Table 2 Determinants of households' WTP for improved water services (Binary Logistic Regression Results). Source Author's computation

Variable	Coefficient (B)	Wald Statistic	p-value	Odds Ratio [Exp(B)]	95% CI for Exp(B)
Household Income	2.014	10.318	0.001 ***	7.490	[2.41, 23.25]
Age of Respondent	0.711	8.670	0.003 ***	2.036	[1.28, 3.24]
Gender (1 = Female)	0.732	4.041	0.044 ****	2.080	[1.02, 4.23]
Marital Status (1 = Unmarried)	-0.572	2.780	0.095 *	0.564	[0.29, 1.10]
Attitude towards water utility (1 = Negative)	-1.640	5.206	0.023 **	0.194	[0.05, 0.79]
Satisfaction with Timing (1 = Dissatisfied)	-1.754	7.927	0.005 ***	0.173	[0.05, 0.60]
Household Water Use (quantity)	0.830	2.795	0.095 *	2.293	[0.87, 6.03]
Primary Water Source (1 = Piped)	0.421	2.145	0.143	1.523	[0.87, 2.67]
Constant	-1.508	3.762	0.052	0.221	[0.05, 1.01]

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Forest Plot of Determinants of Households' Willingness to Pay (WTP)

**Fig. 3** Forest Plot of Determinants of Households' WTP (Odds Ratio and 95% CI)

Household Water Use: The current quantity of water used by the household had a positive and marginally significant influence on WTP ($p < 0.1$). Households that consume more water are more likely to value and be willing to pay for reliable improvements (Odds Ratio = 2.293).

Attitude of Households: This variable had a strong, statistically significant negative impact on WTP ($p < 0.05$). A negative attitude towards the water utility drastically reduced the odds of being willing to pay. The odds ratio of 0.194 implies that households with a negative attitude have 80.6% lower odds of WTP compared to those with a positive or neutral attitude, *ceteris paribus*.

Satisfaction with Time-of-Service Provision: Dissatisfaction with the current timing and reliability of water delivery was also a powerful negative factor ($p < 0.01$). The odds ratio of 0.173 suggests that dissatisfaction with service timing reduces the odds of WTP by 82.7%, highlighting the critical importance of perceived service performance.

The magnitude and significance of the determinants are visually summarized in Fig. 3, a forest plot of the adjusted odds ratios.

This plot visualizes the results of the binary logistic regression (Table 2 in the manuscript), showing the Adjusted Odds Ratio (AOR) for each determinant of Willingness to Pay (WTP), along with the mathematically derived 95% Confidence Intervals (CI).

Key Findings Illustrated:

- Household Income is the strongest positive predictor, increasing the odds of WTP by nearly 7.5 times.
- Institutional Factors (Attitude and Satisfaction with Timing) are powerful negative determinants, severely reducing the odds of WTP (by over 80%).
- Gender (Female) and Age are positive predictors, suggesting women and older respondents have higher odds of WTP.

4.3 Robustness checks and supplementary analyses

To assess robustness, supplementary analyses were conducted. An interaction between income and negative attitude was non-significant ($p = 0.212$), suggesting trust's effect is consistent across income levels. Sensitivity analyses confirmed excluding insignificant variables did not alter core findings.

Economic Implications: The mean WTP offers a tariff benchmark. If the 68.9% of willing households paid 98.5 ETB, potential monthly revenue could be ~250,000 ETB. However, WTP varied by income (lowest quintile: ~45 ETB), underscoring the need for a progressive tariff.

Qualitative Insights on Trust: Analysis of open-ended refusal reasons revealed themes like “Lack of Faith in Improvement” and “Past Betrayal,” illustrating the causal link between experienced institutional failure and the decision not to pay.

5 Discussion

The findings of this study provide robust and detailed evidence that households’ willingness to pay for improved water services in Leku Town is a complex decision influenced by an intricate interplay of economic capacity, demographic characteristics, and, most critically, perceptions of institutional performance and trust. The results strongly support the conceptual framework that WTP is a function of both *abilities* to pay (largely determined by income) and *willingness* to pay (influenced by trust and expectations of service quality).

The dominant positive effect of household income is a consistent finding across contingent valuation studies in Ethiopia and the broader developing world. This finding strongly reinforces the conceptual framework’s proposition that economic capacity (ability to pay) is a fundamental driver of WTP, aligning with prior empirical evidence [7, 26, 76]. The result, showing that higher income vastly increases the likelihood of WTP (Adjusted odds ratio, AOR = 7.49), aligns perfectly with economic theory and prior research. For instance, a study in Sebeta Town, Ethiopia, found income to be the single most important factor affecting WTP [21]. Similarly, research in the Central Rift Valley of Ethiopia [43] and in Jigjiga Town [33] confirmed a strong positive relationship between household income and WTP for improved water services. This underscores that affordability is a primary gatekeeper for participation in cost-sharing models. It implies that without targeted pro-poor measures, such as increasing block tariffs or direct subsidies, tariff increases could exclude the most vulnerable households from accessing improved services, potentially exacerbating existing inequalities and violating the human right to water [61, 69].

The positive association between age and WTP found in this study (AOR = 2.036) offers an interesting point of discussion. This finding contrasts with some studies, such as that by Genanew Bekele [25] in Harar, which found no significant relationship, but aligns with others that suggest older household heads, who may have experienced water problems for longer and potentially have more stable financial situations, place a higher value on reliable water access [35, 66]. This divergence highlights the context-specific nature of demographic influences and suggests that the lived experience of water scarcity over time can be a significant motivator for investment in permanent solutions.

The finding related to gender is particularly noteworthy and carries profound policy implications. The analysis revealed that female respondents were significantly more likely to express WTP than male respondents (AOR for males = 0.522). This creates a vicious cycle: poor water services disproportionately burden women, and their exclusion from planning perpetuates this inequality, leading to services that fail to meet their needs, thereby reinforcing the burden [17, 24]. For example, Taye et al. [65] study in Addis Ababa on solid waste management also found female-headed households to have

a higher WTP. This can be largely explained by the gendered division of labor in water-related activities. In Ethiopia, as in much of the developing world, women and girls are primarily responsible for water collection and management at the household level [27, 62]. They therefore bear the direct brunt of poor water services investing significant time, enduring physical strain, and dealing with the health consequences for their families. Consequently, they experience the negative impacts of inadequate water supply more acutely and are likely to place a higher value on improvements, perceiving a greater net benefit from the proposed service change. This finding strongly suggests that women should be central stakeholders in water planning and decision-making processes, as their preferences more accurately reflect the primary demand for reliable water [54, 64]. Their exclusion not only perpetuates gender inequality but also leads to inefficient and less sustainable water project design.

Perhaps the most critical contribution of this study lies in the powerful negative influence of institutional and performance-related factors. The results demonstrate that a negative attitude towards the water utility decreases the odds of WTP by over 80%, while dissatisfaction with service timing decreases the odds by over 82%. These effects are not just statistically significant but are also economically substantial, nearly as powerful as income in deterring WTP. This provides robust empirical weight to the theoretical argument that technical and financial solutions are insufficient without parallel improvements in governance and institutional trust [5, 52, 60].

The finding on the importance of institutional trust resonates strongly with recent research in other public good domains. Shan et al., [60], in their study on municipal solid waste treatment in Beijing, China, found that institutional trust was a key mediator for WTP, concluding that “ignoring the low level of institutional trust would be detrimental to the promotion of public participation” (p. 1177). Similarly, in the context of water governance, Jibat et al., [36] emphasize the crucial role and interplay of formal and informal institutions, arguing that effective service delivery is underpinned by trust and accountability. The situation in Leku Town appears to be a classic case where past and current service failures, likely characterized by broken promises, unreliable supply, and poor communication, have eroded public trust, a phenomenon documented in other Ethiopian utilities [5, 8]. Households, acting as rational economic agents, are hesitant to commit to higher payments if they doubt the utility’s capability or commitment to deliver on its promises. This creates a vicious cycle: low revenue due to distrust leads to poor service, which further reinforces distrust and suppresses WTP [87]. Breaking this cycle requires addressing these perceptual barriers directly through consistent performance and transparent engagement.

The positive influence of current water use (AOR = 2.293) is intuitive and consistent with demand theory. Households that already consume larger volumes of water, perhaps due to larger household size or higher living standards, stand to benefit more from a reliable and convenient supply and are thus willing to pay more for it. This aligns with findings from studies like Bogale and Urgessa [7] in rural eastern Ethiopia. The marginally significant effect of marital status, with unmarried individuals being less willing to pay, could be linked to differences in perceived future benefits or stability. Married respondents, often with children and planning for a long-term household future, may place a higher value on the long-term health and convenience benefits of improved water, a finding that echoes the stability often associated with older age.

6 Conclusion and recommendations

6.1 Conclusion

This study set out to assess households' willingness to pay for improved water services in Leku Town, a representative secondary Ethiopian town, and to identify the key factors influencing this decision using an integrated analytical framework. The findings lead to several conclusive points.

First, there is a substantial latent demand for better water services in Leku Town, with a significant majority (68.9%) of households expressing willingness to pay. This represents a vital opportunity for the local water utility to enhance its financial sustainability through improved cost-recovery mechanisms. However, a critical paradox exists: realizing this potential revenue requires upfront investment to improve services and rebuild trust. This necessitates initial catalytic financing from local government or development partners to break the cycle. The potential revenue from this willingness could be leveraged for infrastructure upgrades and operational improvements.

Second, the determinants of WTP are multifaceted and extend beyond simple economics. Consistent with global literature, household income is the most powerful positive driver, underscoring the non-negotiable importance of economic capacity. Demographic factors such as age, gender, and marital status also play significant and nuanced roles. The particularly strong preference for improvements among women underscores the need for gender-sensitive water policies.

Third, and most critically, this study provides compelling evidence that technical and economic considerations are insufficient without addressing the fundamental issue of institutional trust. The strong negative effects of a negative household attitude and dissatisfaction with service delivery reveal a significant trust deficit. Households are not merely passive recipients of services; they are rational actors whose financial participation is contingent upon their confidence in the service provider's performance and reliability. In the context of Leku and similar towns, overcoming this trust barrier is not a secondary concern but a prerequisite for unlocking household investment in public goods. Furthermore, securing sustainable financing through improved tariffs and trust is a prerequisite for water utilities to invest in source protection and climate-resilient infrastructure, addressing the environmental dimension of SDG 6.

To sum up, the importance of this research lies in its explicit integration and empirical validation of institutional trust variables alongside traditional socioeconomic factors in the under-researched context of a secondary Ethiopian town. It demonstrates that the sustainability of water services hinges on a dual strategy: enhancing the *ability* of the poor to pay while simultaneously rebuilding the *willingness* of all households to pay by restoring institutional credibility.

6.2 Policy recommendations

Based on the conclusive findings, the study proposes the following targeted recommendations derived directly from the key determinants identified:

1. **Design and Implement a Pro-Poor Tariff Structure:** *Derived from the powerful income finding.* Given the overpowering influence of income, a blanket tariff increase is inequitable. The water utility should design a progressive Increasing Block Tariff (IBT). This should include a low-priced "lifeline" block covering basic needs for

all, with higher per-unit prices for successive consumption blocks. This ensures affordability for the poor while securing higher contributions from those with greater capacity to pay, enabling cross-subsidization.

2. **Launch Transparent, Proactive Communication Campaigns:** *Derived from the institutional trust finding.* Before any tariff change, the water utility must proactively and clearly communicate using simple language and multiple channels how every Birr collected will be directly linked to specific, tangible service improvement plans (e.g., “Your payment will help us repair the pump in your kebele”). This transparency is essential to rebuild credibility and manage expectations.
3. **Ensure Inclusive Governance with Mandated Gender Representation:** *Derived from the gender finding.* Through structures such as community water committees and tariff consultation forums, mandate a minimum representation for women (e.g., 50%) and actively partner with women’s groups. Their inclusion ensures that service design reflects the needs of the primary users and managers of household water, leading to more effective and sustainable projects.
4. **Prioritize Visible Maintenance to Rebuild Trust:** *Derived from the institutional trust/satisfaction finding.* To counteract the trust deficit, the water utility must *demonstrate* commitment through highly visible, well-communicated maintenance actions (e.g., fixing major leaks in specific kebeles). This provides tangible proof that payments lead to improvements and is not merely an obvious technical step, but a core trust-building strategy grounded in the study’s findings.

By adopting this integrated strategy combining equitable financing, tangible service improvements, transparent communication, and inclusive engagement, the Leku Town water utility can effectively address both the *ability* and *willingness* to pay. This holistic approach is the most viable pathway to breaking the debilitating cycle of poor service and low revenue, thereby paving the way for a sustainable, efficient, and equitable water supply system that significantly improves public health, economic well-being, and social cohesion in the community.

6.3 Study limitations and avenues for future research

This study has several limitations that also point to directions for future research. First, the use of a cross-sectional design means that we can establish associations but not definitive causal relationships between the independent variables and WTP. Longitudinal studies tracking changes in WTP following actual service improvements would provide stronger causal evidence. Second, while the sample is representative of Leku Town, the findings are context-specific to secondary towns with similar governance and infrastructure challenges. Caution should be exercised in generalizing results to large metropolitan areas or rural settings. Third, the sampling frame relied on official kebele household lists, which likely exclude recent migrants and informal settlements, potentially biasing the sample toward more established residents. Fourth, wealth indicators were not captured, which could complement income as a measure of economic capacity. Fifth, while interpreting the strong association between institutional perceptions and WTP as indicative of a critical relationship, the potential endogeneity is acknowledged. Attitudes may be shaped by unobserved factors that also influence WTP. Future research using panel data or experimental designs could better establish causality. Sixth, the CVM is inherently hypothetical, and respondents stated that WTP may not perfectly predict

their actual payment behavior. Future research could employ experimental methods, such as randomized controlled trials of new tariff systems, to observe real-world behavior. Seventh, trust was measured as a general attitude; future studies could develop and employ more detailed, multi-dimensional scales to measure different facets of institutional trust (e.g., competence, integrity, benevolence). Finally, while the high model fit is encouraging, the predictive power of institutional variables may be context-specific; findings should be validated through comparative studies across many secondary cities.

Supplementary Information

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Supplementary Material 1

Supplementary Material 2

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Author contributions

The author solely conceived, designed, conducted, analyzed, and wrote the manuscript.

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Data availability

The anonymized dataset generated and analyzed during the current study is not publicly available due to confidentiality agreements with participants and conditions of the ethical approval, but is available from the corresponding author upon reasonable request for academic verification purposes. The full survey questionnaire is provided as Supplementary Information.

Declarations

Ethical approval

This study received ethical clearance from the Institutional Review Board (IRB) of the College of Business and Economics at Hawassa University, Number CBE-RTT-50/2024. All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Consent to participate

Informed consent was obtained from all individual participants before data collection. Participants were informed of the study's purpose, their voluntary participation, the right to withdraw at any time, and the confidentiality of their responses. Because the study focused on the household level, individuals under 16 were excluded; therefore, parental or guardian consent was not required.

Consent for publication

Consent to Publish declaration: not applicable.

Competing interests

The authors declare no competing interests.

Assistance of AI tools

AI-assisted tools were used for linguistic editing and refinement. The author assumes full responsibility for the scholarly content, analysis, and conclusions.

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